

Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence

Erik Brynjolfsson* Bharat Chandar† Ruyu Chen‡§¶

August 26, 2025

Abstract

This paper examines changes in the labor market for occupations exposed to generative artificial intelligence using high-frequency administrative data from the largest payroll software provider in the United States. We present six facts that characterize these shifts. We find that since the widespread adoption of generative AI, early-career workers (ages 22-25) in the most AI-exposed occupations have experienced a 13 percent relative decline in employment even after controlling for firm-level shocks. In contrast, employment for workers in less exposed fields and more experienced workers in the same occupations has remained stable or continued to grow. We also find that adjustments occur primarily through employment rather than compensation. Furthermore, employment declines are concentrated in occupations where AI is more likely to *automate*, rather than *augment*, human labor. Our results are robust to alternative explanations, such as excluding technology-related firms and excluding occupations amenable to remote work. These six facts provide early, large-scale evidence consistent with the hypothesis that the AI revolution is beginning to have a significant and disproportionate impact on entry-level workers in the American labor market.

*Stanford University and NBER; erikb@stanford.edu

†Stanford University; chandarb@stanford.edu

‡Stanford University; ruyuchen@stanford.edu

§Thanks to Nick Bloom, Joshua Gans, David Autor, Daniel Rock, Fei-Fei Li, Frank Li, Christina Langer, Sarah Bana, Cody Cook, Chris Forman, Andrew Wang, Brad Ross, Omeed Maghzian, Basil Halperin, Jiaxin Pei, Phil Trammell, Eric Bergman and participants at the Stanford Digital Economy Lab workshop for helpful feedback. We are grateful to ADP for access to the data and the Stanford Digital Economy Lab for financial support. All errors are our own.

¶Latest version: <https://digitaleconomy.stanford.edu/publications/canaries-in-the-coal-mine/>

Figure 3 shows these patterns hold more generally across professions. The top left plot shows a divergence in employment outcomes for more and less exposed occupations for workers aged 22-25, with more exposed occupations experiencing declining employment. For older age groups, we find much less marked differences in employment growth across AI exposure quintiles.

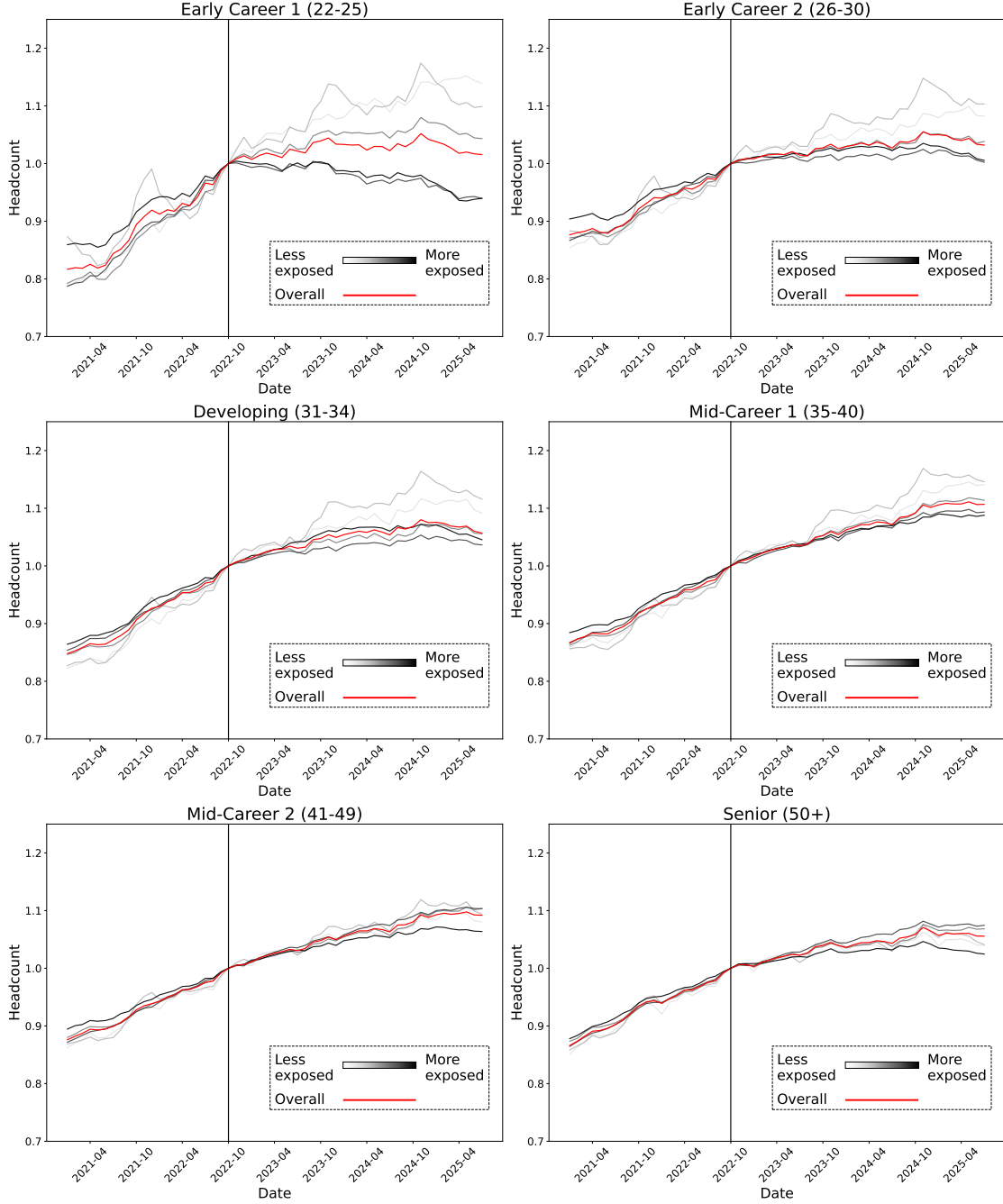


Figure 3: $GPT-4 \beta$. Employment changes by age and exposure quintile using measures from Eloundou et al. (2024). Exposure quintiles are defined based on the $GPT-4 \beta$ measures. Darker lines are more exposed quintiles. The red line shows the overall trend pooling across quintiles.